

Synthesis Characterisation Of Iodocoumarin And Its Evaluation As An Antimicrobial Agent Against Selected Pathogens.

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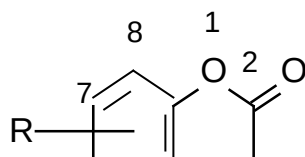
Abstract-

Coumarins are among the largest groups of secondary natural products found frequently in higher plants and less commonly in microorganisms. The 2H-pyran-2-one structural unit appears more often in naturally occurring products that often exhibit remarkable biological profiles. In the present study, we have synthesized some new iodo coumarins by condensation of o-hydroxyiodo substituted acetophenones or o-hydroxyiodo-substituted benzaldehydes with different organic acids and fused sodium acetate by irradiating under microwaves led to the formation of iodo substituted coumarins. The purity of compounds was checked by TLC. Melting-point and yields of the product were obtained. Structures of all compounds were confirmed by IR, ¹HNMR and mass spectroscopy. Further the antimicrobial activities of synthesized iodocoumarins on plant and animal including human pathogenic bacteria namely (a) *X.citrii* causing citrus canker (b) *S.aureus* causing broad range of skin infection and antifungal activities on plant pathogenic fungi (c) *C.personatum* -leaf spot pathogen of groundnut and (d) *H.torulolum*- a leaf blight and fruit rot pathogen of Banana were evaluated. Most of the synthesized compounds showed excellent antimicrobial activities with clear Zone of inhibition. Iodo derivatives were found to be more potent as compared to chloro derivatives. Thus the newly synthesized iodocoumarins and its further development will lead to promising antimicrobial agents.

Keywords : iodocoumarin, microwave irradiation, antibacterial, antifungal.

INTRODUCTION:

Coumarin, one of the secondary metabolites of plants and microorganism encompassing of simple heterocyclic structure of oxygen, are well-known for their antimicrobial, anticoagulant, anti-inflammatory, antihelmenthic, antifeedent, hypnotic, analgesic properties. Fused heterocyclic coumarins have diverse biological importance in treatment of vitiligo, psoriasis and coronary diseases.



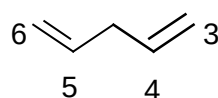


Fig1. Molecular structure of coumarin.

Coumarins are among the largest groups of secondary natural products found frequently in higher plants and less commonly in microorganisms [1]. The 2*H*-pyran-2-one structural unit appears more often in naturally occurring products [2] that often exhibit remarkable biological profiles.

As it has a sweet aroma which resembles the smell of new mown hay and hence coumarins are used in scent industry. As coumarin is an antifeedent it is assumed as the substance produced by the plants as chemical defense to put off predation [3, 4]. Coumarins are a class of lactones which consist of a benzene ring fused with α -pyrone ring and fundamentally hold an electron rich conjugated system with good charge transfer properties [5]. The flexibility, adaptability and straightforwardness of coumarins make them an attractive starting material for a wide range of applications [6-8]. Coumarins and its derivatives are used in perfumery, cosmetics, industrial additives, aroma enhancers in tobaccos and alcoholic drinks [9, 10]. Nevertheless the other role played by coumarins which is of high significance is in organic chemistry, medicinal chemistry and study of natural products [11, 12]. The separation, extraction, purification, evaluation and synthetic studies of coumarins have turned out to be a tremendously attractive and speedily budding subject for researchers [13, 14]. the coumarin compounds established an important place in the field of medicinal chemistry [15, 16] and thoroughly studied as a bioactive agent [17]. Furthermore, coumarins containing compounds have high pharmacological activity, less toxicity, less side effects, high bioavailability, high spectrum, and all other properties that a drug must have to be widely accepted and used [18]. Mode of action of coumarin is closely studied and many more efforts are made in developing drugs which contain coumarin. Some examples can be given as coumarin based anticoagulants, antioxidant [19], antitumor [20-22] pain killer, pain reliever, anti-neurodegenerative, anti-diabetic, and anti-inflammatory agents [23]. It can be used as active antimicrobial agent that includes antibacterial, antifungal, antiviral, antihelmenthic, antiparasitic properties [24].

PRESENT WORK MATERIALS AND METHOD USED

SYNTHESIS OF IODO HYDROXYACETOPHENONES AND IODOHYDROXYBENZALDEHYDE USING IODINE AND IODIC ACID:

For iodination of acetophenones many iodinating reagents are known like iodine monochloride but it is hazardous. Iodine and ammonia but it forms ammonium salts and iodination remain incomplete. Pyridine-iodinemonochloride complex has been utilized as an iodinating agent but isolation of product is not easy. Iodination is also performed using iodine and NaOH, iodine, KI and KIO₃ in acetic acid. But most convenient and effective iodinating agent which has been reported by B. S. Dawane and Y. B. Vibhute [25] for iodination of hydroxyacetophenones is iodine and iodic acid. It is a very simple method where isolation of product is easy and yield is very high. A mixture of

2'-hydroxyacetophenone (0.001 mol) and iodine (0.001 or 0.002 mol) dissolved in ethanol (30 ml) was added to iodic acid (0.001 or 0.002 mol) dissolved in water (1 ml) and stirred for 0.5 h at 35-40 C. It was further stirred for 1 hour and diluted with cold water. The resulting solid products were washed successively with saturated sodium bisulfite solution and cold water and filtered, dried and crystallized from ethanol. The purity of compounds was checked by TLC. The m.p. and percent yields of the products were recorded in Table1. Similarly substituted benzaldehyde is also iodinated following same procedure.

Preparation of 3-phenyl-6,8-diiodo-7-hydroxycoumarin (3j):

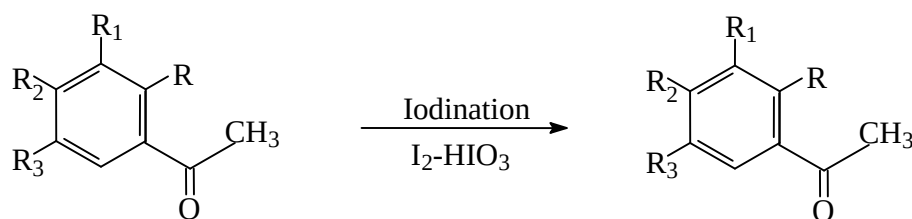
A mixture of 2,4-dihydroxy, 3,5-diiodomobenzaldehyde (0.001 mole), phenylacetic acid (0.001 mole) and fused sodium acetate (0.001 mole) in acetic anhydride (2ml) was irradiated in a microwave oven for 1.5 min. with an interruption of 4-5 min. (30 sec. x 3 times). Completion of the reaction was monitored by TLC. The reaction mixture was allowed to cool at room temperature, poured onto ice or ice-cold water with constant stirring and kept overnight. The solid separated was filtered, washed, dried and recrystallised from aqueous acetic acid. All the synthesised compounds were characterised on the basis of their physical, analytical data (Table-3) and spectral data (IR, ^1H NMR and Mass spectroscopy). All the other compounds were prepared by similar procedure.

Characteristic test:

The ferric chloride test is used to determine the presence of phenols in given sample. Aromatic hydroxides like phenols form characteristic coloured iron complexes when treated with aq. FeCl_3 solution. Phenols gives violet colour, resorcinol gives catechol green color. Enols, hydroximic acids, oximes etc gives positive results as well. The procedure can be given as follows. The sample is dissolved in water or ethanol and few drops of aq FeCl_3 solution is added. The formation of red, blue, green or purple coloration indicates presence of phenols.

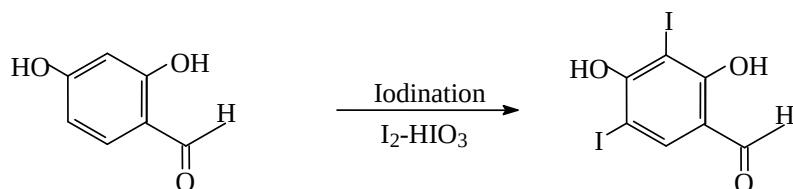
Characteristic tests for coumarins :

When the synthesized iodocoumarins were treated with aq. Ferric chloride no coloration is observed. Hence all the coumarins did not gave test for Ar-OH group with aq. FeCl_3 , and the test for Ar-CO- CH_3 group.



Hydroxyacetophenones (a - b)

Iodohydroxyacetophenones 1a – 1b



2,4-dihydroxybenzaldehyde

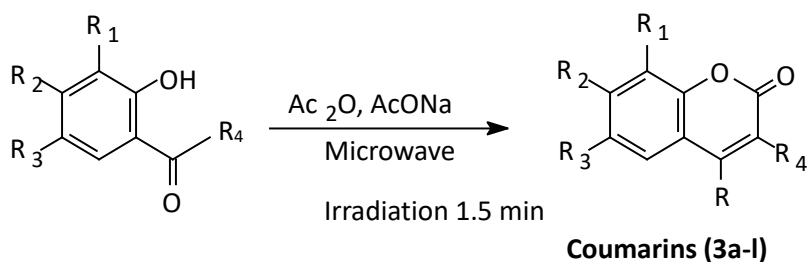
2,4-dihydroxy-3,5-diiodobenzaldehyde

(1c)

SCHEME1: Iodination of hydroxyacetophenones and dihydroxybenzaldehyde using iodinating agent I_2-HIO_3 .

Table no.1. Various substituents at R,R₁,R₂,R₃ position with % yield and melting point of 1a, 1b and 1c:

Compd.	R	R ₁	R ₂	R ₃	Yield(%)	M.P(^o C)
1	OH	I	CH ₃	Cl	86%	74 ^o C
1b	OH	I	H	I	79%	128 ^o C
1c	OH	I	OH	I	86%	269 ^o C



SCHEME2: Synthesis of coumarins using acetic anhydride and sodium acetate by microwave irradiation for 1.5 min. **Table2.** List of synthesized Coumarins with various substituents at R, R₁, R₂, R₃ and R₄ positions (3a-l):

BIOLOGICAL ACTIVITY:

Treatment of diseases with chemical substances has been known since the fifteenth century. Chemical agents not only provide the structure basis and energy supply of living organism but also regulate their functional activities. The interaction between potent chemical and living system contribute to the understanding of the life process and provide effective method for the treatment, prevention and diagnosis of many diseases. Fighting against diseases with drugs is the endless struggle. The field and scope of medicine is too vast.

Comp.	R	R ₁	R ₂	R ₃	R ₄
3a	CH ₃	I	H	I	H
3b	CH ₃	I	H	I	Ph
3c	CH ₃	I	H	I	NHCOCH ₃
3d	CH ₃	I	H	I	NHCOPh
3e	CH ₃	I	CH ₃	Cl	H
3f	CH ₃	I	CH ₃	Cl	Ph
3g	CH ₃	I	CH ₃	Cl	NHCOCH ₃
3h	CH ₃	I	CH ₃	Cl	NHCOPh
3i	H	I	OH	I	H
3j	H	I	OH	I	Ph
3k	H	I	OH	I	NHCOCH ₃
3l	H	I	OH	I	NHCOPh

PRESENT WORK AND METHOD USED:

Antibacterial Activity

The information regarding the various species of bacteria used to carry out screening is given below: Active Cultures of the *Xanthomonas citri* (Xc), *Escherichia coli* (Ec), *Staphylococcus aureus* (Sa) were obtained from department of biotechnology, Yeshwant Mahavidyalaya, Nanded (M.S.) sub culturing was done after every fifteen days. In all experiments the cultures were incubated at 27°C in the laboratory.

a) *Xanthomonas citri* (Xc)

Xanthomonas citri is gram negative, motile by single polar flagellum, yellow pigmented plant pathogen of the pseudomonodaceae family.

b) *Staphylococcus aureus* (Sa)

Staphylococcus aureus is a gram positive non-motile bacterium which is coccal shaped and arranged in colonies or groups. It is commonly found in the nose, respiratory tract and is commonly found in the respiratory tract, nose and on skin of humans and animals. It is pathogenic in nature. Food poisoning, respiratory infections such as sinusitis and skin infection like abscesses are caused by *S.aureus*. *S.aureus* generates many toxins which add to increase the pathogenicity of the bacteria by raising its capability to attack and damage the host.

Antifungal Activity:

In the present investigation, the antifungal activity of newly synthesised compounds 3a-3l coumarins against some selected plant pathogens was carried out. The pathogens used were *Cercosporidium personatum* (Cp) and *Helminthosporium torulosum* (Ht).

Cultures of the *Cercosporidium personatum* (Cp) - a leaf spot pathogen of groundnut and *Helminthosporium torulosum* (Ht) - leaf blight and fruit rot pathogen of Banana, were obtained from Botany Research Laboratory, Yeshwant Mahavidyalaya, Nanded (M.S.). The cultures were maintained on Potato Dextrose Agar (PDA) and sub culturing was done after every fifteen days. In all experiments the cultures were incubated as 27°C in the laboratory.

c) *Cercosporidium personatum* (Cp)

Cercosporidium personatum (Cp) is a fungus which causes leaf spot disease of groundnut plant. It is also known as Tikka disease of groundnut. In India it causes 15-60% loss in yield.

d) *Helminthosporium torulosum* (Ht)

Helminthosporium torulosum (Ht) is a fungus which causes leaf blight and fruit rot disease of Banana plant. It is characterized by blackening of banana at the ends also known as crown rot. It is the major cause of post-harvest loss which thereby affects the economy.

EXPERIMENTAL:

Antibacterial activities:

The well diffusion method was used for the determination of antibacterial activity of the newly synthesized compounds. All the synthesized compounds were recrystallized before use. The

antibacterial and antifungal activities of all the synthesized compounds were studied in DMSO and DMF respectively. Agar well diffusion method was used.

Preparation of the test compound: The synthesized compounds were dissolved in DMSO and DMF at concentration of 2 µg /1 ml. The nutrient agar plates were prepared for growth of the bacterial species. Microorganisms obtained from Department of Biotechnology, Yeshwant Mahavidyalaya, Nanded were seeded on nutrient agar plates. The samples were loaded on the seeded nutrient agar plate. These plates were kept for diffusion in refrigerator for 10-15 min for diffusion. Later after diffusion, the plates were incubated at 37⁰ C for 24 hrs.

The zones of inhibition were measured in millimeters and recorded. The inhibition diameter were recorded between 10-30 mm. Almost all complexes have significant antibacterial activity at 2 µg /1 ml against bacteria using agar well methods. The growth was expressed as zone of inhibition and is given in the table 4 and 5.

Antifungal activities by Spore Germination:

Spore suspensions of Cultures of the *Cercosporidium personatum* (Cp) - a leaf spot pathogen of groundnut and *Helminthosporium torulosum* (Ht) was prepared by adding sterile water to 8 days old PDA slant culture. In every case spore suspension was standardised to contain 1x10⁵ spore/ml using haemocytometer. All the treatment/ sampling were done in triplicate and results have been presented after repeating the experiment.

The spore concentration was measured by Haemocytometer with Naubauer Counting Chamber [26]. Hanging drop slides were prepared for observation from the suspension and incubated in Petri plates lined with moist blotters. Germination % was calculated by observing a minimum of 200 spores from different microscopic field.

The technique or procedure that is followed for antifungal activities is hanging drop technique [27]. This mixture was subjected to observation by spore germination method. Solution without compound was used as control. Hence, experiments were performed to study the effect of the compounds on the spore germination. The spore germination with respect to time were recorded and subsequently the % germination are given in Table 5.

RESULTS:

Table 3. Analytical and Physical data of Coumarins (3a-l) :

Compd.	MOL. FORMULA	Yield (%)	M.P. (°C)	Mol. wt
3 ^o	C ₁₀ H ₆ O ₂ I ₂	78	158	410
3b	C ₁₆ H ₁₀ O ₂ I ₂	76	144	486

3c	C ₁₂ H ₉ O ₃ NI ₂	78	142	467
3d	C ₁₇ H ₁₁ O ₃ NI ₂	74	86	529
3e	C ₁₁ H ₈ O ₂ ICl	80	185	333.5
3f	C ₁₇ H ₁₂ O ₂ ICl	79	143	409.5
3g	C ₁₃ H ₁₁ O ₃ NI ₂	80	137	390.5
3h	C ₁₈ H ₁₃ O ₃ NI ₂	76	196	452.5
3i	C ₉ H ₄ O ₃ I ₂	76	120	412
3j	C ₁₅ H ₈ O ₃ I ₂	75	138	488
3k	C ₁₁ H ₇ O ₄ NI ₂	78	142	469
3l	C ₁₆ H ₉ O ₄ NI ₂	76	138	531

All synthesized compounds were recrystallised from aqueous acetic acid.

SPECTRAL DATA:

IR spectra of the compounds were scanned on Perkin-Elmer FTIR spectrometer. ¹H NMR spectra were recorded on a Gemini 200 MHz instrument using TMS as an internal standard (Chemical shift are given in δ ppm). The mass spectra (MS) were recorded on VG 7070H mass spectrometer using ionisation energy of 70eV.

IR Spectra:

The IR spectra of the coumarins and 4-methylcoumarins showed characteristic absorption band at 1700-1730 cm⁻¹ due to C-O stretch of lactone, beside this the disappearance of band at 1630-1640 cm⁻¹ or 1650-1670 cm⁻¹ suggest that C=O stretching due to COCH₃ group is not present. In the compounds where -OH group is present at 7-position, a band at 3200 cm⁻¹ is observed due to -OH stretching vibration. While the 2'-OH of acetophenone is disappeared. At near 1600 cm⁻¹, C = C stretching vibrations appeared. 2875-2950 cm⁻¹ (methyl C-H stretching), $\approx$ 3025 cm⁻¹ (aromatic C-H) and at 800 C-Cl, $\approx$ 700 cm⁻¹ C-Br stretching are observed.

¹H NMR Spectra:

The ¹H NMR spectra of the compounds synthesised showed characteristic singlet peaks at δ 2.30-2.45 due to Ar-CH₃. A singlet is observed near at δ 6.2-7.2 due to one Hydrogen at C-3 position of the coumarin (when it is unsubstituted) and aromatic proton peak at δ 7.4-7.5 is observed for C-5 proton. These characteristic peaks observed for the IR and ¹H NMR of the coumarins are in agreement with those observed earlier.

Mass Spectra:

Mass spectroscopy is an important tool for determining the structure of the synthesised iodocoumarins.

Compd. 3j :

IR : ν max : 3248, 3040, 2864, 1705, 690

^1H NMR (CDCl_3) : δ 8.23 (s, 5H), 2.75 (s, 1H, 4-Ar-H),
7.48 (d, 1H, 5-Ar-H), 6.7 (s, 1H, 7-Ar-OH).

Mass m/z : 201(M^+), 173,191

Compd. 3a :

IR : ν max : 2950, 1600, 1710,590

^1H NMR (CDCl_3) : δ 2.38 (s, 3H, Ar- CH_3), 7.25 (s, 1H, 3-Ar-H),
7.53 (s, 1H, 5-Ar-H), 7.8 (s, 1H, 7-Ar-H).

Mass m/z : 208,180,163

Compd. 3e :

IR : ν max : 3228,2870,1705,800

^1H NMR (CDCl_3) : δ 2.48 (s, 3H, C_4 -Ar- CH_3),
2.36 (s, 3H, C_7 -Ar- CH_3),
6.22 (s, 1H, C_3 -Ar-H),
7.48 (d, 1H, C_5 -Ar-H).

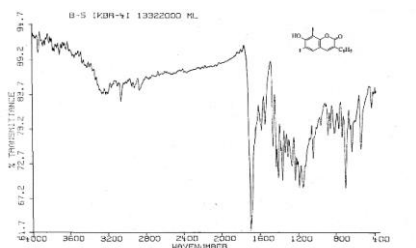
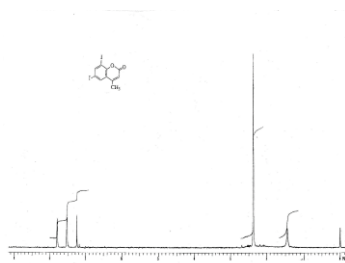


Figure 2: IR spectra of 7-hydroxy-8-methyl-3-phenyl-6-iodocoumarin

Compd. 3j :



IR : $\nu_{\max}$: 3248, 3040, 2864, 1705, 690

Figure 3: ^1H NMR spectra of 4-methyl-6,8-diiodocoumarin

Compd. 3a :

^1H NMR (CDCl_3) : δ 2.38 (s, 3H, Ar- CH_3), 7.25 (s, 1H, 3-Ar-H),
7.53 (s, 1H, 5-Ar-H) , 7.8 (s, 1H, 7-Ar-H).

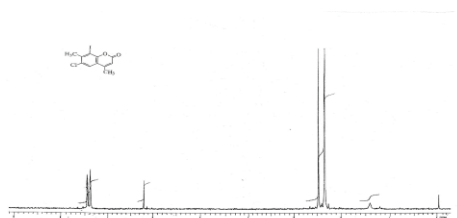


Figure 4: ^1H NMR spectra of 6-chloro-4,7-dimethyl-8-iodocoumarin

Compd. 3e :

^1H NMR (CDCl_3) : δ 2.48 (s, 3H, C4-Ar- CH_3),
2.36 (s, 3H, C7-Ar- CH_3),
6.22 (s, 1H, C3-Ar-H),
7.48 (d, 1H, C5-Ar-H).

RESULTS OF BIOLOGICAL ACTIVITIES:

The results obtained are tabulated in the tables 4 and 5.

The compounds 3a, 3b and 3c showed excellent activities of growth inhibition on bacteria *S.aureus*. Whereas good activities were exhibited by 3d, 3e and 3f compounds which conclude that the compounds 3a to 3f possess high antibacterial activity and compounds 3g to 3l exhibits comparatively

less activity compared to 3a-3f. Similarly, for antifungal activity the hanging drop method was employed which counts the rate of fungal spore germination.

The Iodo derivatives were found to be more potent as compared to its chloro derivatives also the activities increases with introduction of hydroxyl group at R₂ position. From the results, inference is drawn that the number of hydroxy groups imparts toxicity against pathogens. The order of toxicity against pathogens also depends on number and type of halogen substitutes. Besides this, the presence of methoxy groups in the ring also enhances their toxicity.



Figure 5: shows Zone of inhibitions for antibacterial activity by well diffusion method by comp C₁ C₂ and C₃ on bacterial colonies of pathogenic strains of *S.aureus*. An excellent antibacterial activity is shown by the compound. Hence, we say that these compounds are potentially very active antibacterial agents

Figure 6: Shows Zone of inhibitions for antibacterial activity by well diffusion method by comp C₄ C₅ and C₆ on bacterial colonies of pathogenic strains of *X.citrii*. A good Antibacterial activity is shown by the compounds. Hence, we say that these compounds are active antibacterial agents

Table 4: Antibacterial Activities of Coumarins on pathogenic bacteria (3a-l)

Compd.*		Zone of inhibition (mm)
	Compd.	Bacteria

	Code	Sa	Xc
3a	C ₁	19	09
3b	C ₂	18	08
3c	C ₃	12	13
3d	C ₄	16	12
3e	C ₅	06	18
3f	C ₆	10	20
3g	C ₇	08	08
3h	C ₈	15	10
3i	C ₉	14	13
3j	C ₁₀	11	12
3k	C ₁₁	12	11
3l	C ₁₂	09	09
Control	-		
Tetracyclin		20	20

*(Entry as taken in Table 1)

Table5: Antifungal Activities of Coumarins on plant pathogenic fungi (3a-l)

Compd. (Entry as taken in Table 1)	Compd. Code	% germination	
		(after 1 hr.) - Fungi	
		Cp	Ht
3a	C ₁	26	31
3b	C ₂	39	35
3c	C ₃	30	32
3d	C ₄	33	29
3e	C ₅	32	35

3f	C ₆	38	36
3g	C ₇	32	28
3h	C ₈	29	24
3i	C ₉	42	32
3j	C ₁₀	31	41
3k	C ₁₁	26	33
3l	C ₁₂	34	18
None		38	44

CONCLUSION

The twelve synthesized iodo coumarins (3a-3l) were assessed for their antibacterial and antifungal activities against selected plant and human pathogens. For the antibacterial studies human pathogenic bacterial strains *Staphalococcus aureus* and plant pathogenic bacterial strains *Xanthomonas citrii* were selected. Infections include pimples boils, impetigo, cellulites, folliculitis, pneumonia, meningitis; endocarditic, toxic shock syndrome, bacteremia and sepsis etc caused by *S.aureus* whereas the citrus cancer disease of citrus plants is caused by *X.citrii*. The standard drug used was tetracycline and the zone obtained was measured in mm. And for the antifungal activity, the fungi *Cercosporidium personatum* and *Helminthosporium torulosum* were selected and the hanging drop method was employed for antifungal activities by measuring the rate of spore germination against time after every 15 minutes.

Compounds 3a to 3l were coded as C₁ to C₁₂ respectively. Compounds which are coded as C₁, C₂, C₄ and C₈ showed excellent antibacterial activity against *S.aureus*. The zone of inhibition obtained for these compounds is 19mm, 18mm, 16mm and 15mm respectively when compared against the standard drug tetracycline whose zone of inhibition obtained was 20 mm. The zone of inhibition which were obtained for C₁ -C₆ compounds were 17mm, 15mm, 16mm, 24mm, 25mm, 25 mm respectively when compared against tetracycline (20 mm). Compounds 3d, 3e, and 3f which are coded as C₄(16mm), C₅(18mm), C₆(20mm) shows excellent activity against plant pathogenic bacterial strain of *X. citrii*. (fig 3.5). Whereas for antifungal studies we can interpret the data that the Compounds 3b, 3f and 3i which are coded as C₂, C₆, C₉ shows excellent activity against fungus *C.personata* while Compounds 3b,3e, 3f and 3j which are coded as C₂, C₅, C₆ and C₁₀ shows excellent activity against fungus *H. torulosum*. The antifungal activity is assessed by the rate of germination of fungal spore. The control used for antifungal activity is DMF solution and

comparison is made with this and can be concluded that slower the rate of germination of spore active is the compound.

From the results, it is clear that almost all compounds showed the excellent antibacterial and antifungal activities. However, variation in their toxicity against the different bacteria and fungi was observed. It may be attributed to different substitutions and functional groups attached to the basic nucleus. However the functional group and their position impart toxic effect against bacteria and fungi. From these results, inference is drawn that the number of hydroxy groups imparts toxicity against pathogens. The order of toxicity against pathogens is also depend on number and type of halogen substitutes like the diiodo derivatives are more potent to chloro iodo derivatives of coumarin. Beside this, the presence of methoxy groups in the ring also enhances their toxicity.

So many of the pathogens used have been reported as resistant against common antibiotics and fungicides. However, it is clear from the present results that preliminary studies showed their good inhibitory properties therefore further studies should be done in order to have these compounds as bactericides and fungicides. Furthermore, the method used for the synthesis of these iodocoumarins is simple, environmentally safe and eco-friendly which causes very little harm to the surrounding.

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